

From Minerals to Megawatts: Capacity Planning in large Scale Energy System Models

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ESOMs: The Status Quo and Methodological Gap

Evolution of ESOMs: sophisticated, multi-sectoral instruments (coupling heat, transport, and industry)^{1,2}.

Traditional Focus: determining the cost-optimal technology mix.

The Core Output: *what, where, and when* to build capacity.

The Methodological Dilemma: Traditional models lack the capability to answer questions beyond their traditional focus (e.g., regarding supply risks or strategic independence).

The Gap: To date, no large-scale model possesses the capability to fully endogenize this *energy-material nexus*³.

¹Löffler et al. (2017) *Designing a model for the global energy system*

²Brown et al. (2018) *PyPSA: Python for Power System Analysis*

³Schulze et al. (2024) *Overcoming the challenges of assessing the global raw material demand of future energy systems*

A Paradigm Shift: From Minerals to Megawatts

Proposed Solution: A Fundamental Redesign of ESOMs

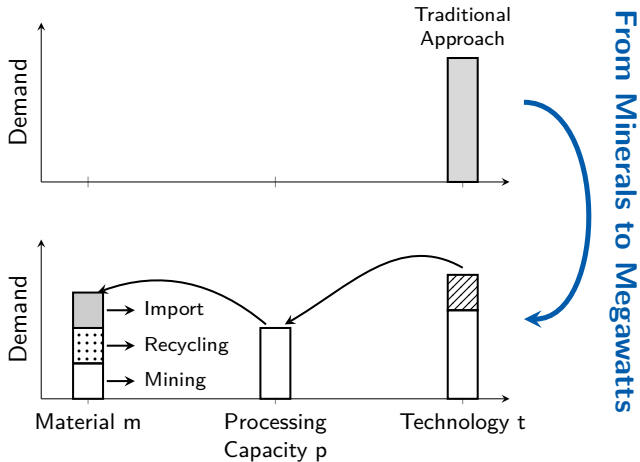
We shift the analytical focus from a purely capacity-driven perspective to a **holistic supply chain orientation**.

Core Objective: *To understand and quantify cost-optimal capacity planning across the entire supply chain.*

Research Questions

- How can typical ESOMs be enabled to design capacities for supply chains of clean technologies?
- What are the cost-optimal capacities and trade-offs along the supply chains of clean technologies between domestic processing and imports?

Framework Design



From Minerals to Megawatts

Linking technology t to processing p

Core Concept: *Final capacity installation determines required processing capacity and processed output.*

Constraint Focus	Mathematical Formulation
Linking supply to final deployment	$\hat{\pi}_{y,\ell,k}^{\text{new}} = \hat{Q}_{y,\ell,k,i=\text{final}}^{\text{supply}} \quad (1)$
Composition of supply for each stage i	$\hat{Q}_{y,\ell,k,i}^{\text{supply}} = \hat{q}_{y,\ell,k,i}^{\text{produced}} + \hat{q}_{y,\ell,k,i}^{\text{imported}} \quad (2)$
Linking subsequent stages until final stage is reached	$\hat{Q}_{y,\ell,k,i}^{\text{supply}} \geq \sum_{k',i'} \hat{q}_{y,\ell,k',i'}^{\text{produced}} \times \alpha_{k',i' \rightarrow k,i} \quad (3)$
Production is limited by facility size	$\hat{q}_{y,\ell,k,i}^{\text{produced}} \leq \hat{p}_{y,\ell,k,i}^{\text{proc}} \quad (4)$
Energy consumption of processing facilities	$E_{y,\ell,k}^{\text{proc}} = \sum_i \hat{q}_{y,\ell,k,i}^{\text{produced}} \times e_{\ell,k,i}^{\text{need}} \quad (6)$

Indices: year y , location ℓ , technology k , processing stage i

Linking processing p to material m

Core Concept: *Processing endogenizes material demand and determines cost-optimal sourcing.*

Constraint Focus	Mathematical Formulation
Determine material demand via μ	$D_{y,r}^{\text{mat}} = \sum_{\ell,k,i} \hat{q}_{y,\ell,k,i}^{\text{produced}} \times \mu_{k,i,r} \quad (7)$
Material sourcing options	$D_{y,r}^{\text{mat}} = \hat{m}_{y,r}^{\text{imp}} + \hat{m}_{y,r}^{\text{mine}} + \hat{m}_{y,r}^{\text{rec}} \quad (8)$
Primary material availability limit	$\hat{m}_{y,r}^{\text{mine}} \leq \frac{M_{y,r}^{\text{avail}} \times s_{y,r}^{\text{ET}}}{c_{y,r}^{\text{mine}}} \quad (9)$
Steady mining growth and decline	$0.70 \times \hat{m}_{y-1,r}^{\text{mine}} \leq \hat{m}_{y,r}^{\text{mine}} \leq 1.30 \times \hat{m}_{y-1,r}^{\text{mine}} \quad (10)$
Secondary sourcing via $\hat{q}_{y,\ell,k}^{\text{rec}}$	$\hat{m}_{y,r}^{\text{rec}} = \sum_{\ell,k} \hat{q}_{y,\ell,k}^{\text{rec}} \times \frac{\phi_{k,r}}{f_{\ell,k}^{\text{scrap}}} \times \eta_{k,r}^{\text{rec}} \quad (11)$

Indices: year y , location ℓ , technology k , processing stage i , raw material r

Retrieving Recycled Materials

Core Concept: *Decommissioned capacity is re-introduced into the system as scrap, closing the loop, serving as source for recycled materials.*

Constraint Focus	Mathematical Formulation
Capacity balance tracking decommissioning	$\pi_{y,\ell,k} = \pi_{y-1,\ell,k} + \hat{\pi}_{y,\ell,k}^{\text{new}} - \hat{\pi}_{y,\ell,k}^{\text{dec}} \quad (12)$
Scrap generation via scrap content $f_{\ell,k}^{\text{scrap}}$	$\hat{q}_{y,\ell,k}^{\text{scrap,dec}} = f_{\ell,k}^{\text{scrap}} \times \hat{\pi}_{y,\ell,k}^{\text{dec}} \quad (13)$
Total scrap is reduced by recycled mass $\hat{q}_{y,\ell,k}^{\text{rec}}$	$\hat{q}_{y,\ell,k}^{\text{scrap,tot}} = \hat{q}_{y-1,\ell,k}^{\text{scrap,tot}} + \hat{q}_{y,\ell,k}^{\text{scrap,dec}} - \hat{q}_{y,\ell,k}^{\text{rec}} \quad (14)$
Different options for recycling wind scrap	$\hat{q}_{y,\ell,k}^{\text{rec}} = \hat{q}_{y,\ell,k}^{\text{rec,mag}} + \hat{q}_{y,\ell,k}^{\text{rec,bulk}} \quad (15)$

Indices: year y , location ℓ , technology k

Example: Economies of scale for the recycling sector

- 1 **Cumulative Output Tracking:** Track historical experience via past recycled quantities.

$$\hat{q}_{y,\ell,k}^{\text{cum,rec}} = \sum_{y'=y_0}^y \hat{q}_{y',\ell,k}^{\text{rec}}$$

- 2 **Stage Selection via Binary Variable:** A binary variable $\delta_{y,\ell,k,n}$ is activated when the cumulative output crosses a defined threshold ($\hat{Q}_{\ell,k,n}^{\text{step}}$), locking the sector into a specific cost-reduction stage n .
- 3 **Big-M Linearization:** The auxiliary variable $\hat{z}_{y,\ell,k,n}$ represents the linearized product of the recycling volume and the active stage ($\hat{q}_{y,\ell,k}^{\text{rec}} \cdot \delta_{y,\ell,k,n}$).
- 4 **Cost Reduction:** The total cost reduction is given by the active stage discount.

$$\Xi_{y,\ell,k}^{\text{red}} = \sum_n P_{\ell,k,n}^{\text{red}} \hat{z}_{y,\ell,k,n}$$

Scenario Constraints: NZIA & CRMA

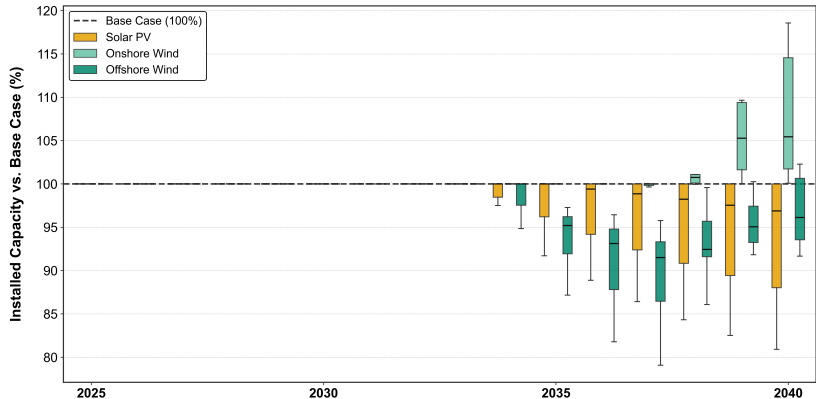
■ Net-Zero Industry Act:

$$\hat{q}_{y,\ell,k,i}^{\text{flow,dom}} + \hat{q}_{y,\ell,k,i}^{\text{stockout,dom}} \geq 0.40 \times \hat{Q}_{y,\ell,k,i}^{\text{supply}} \quad (1)$$

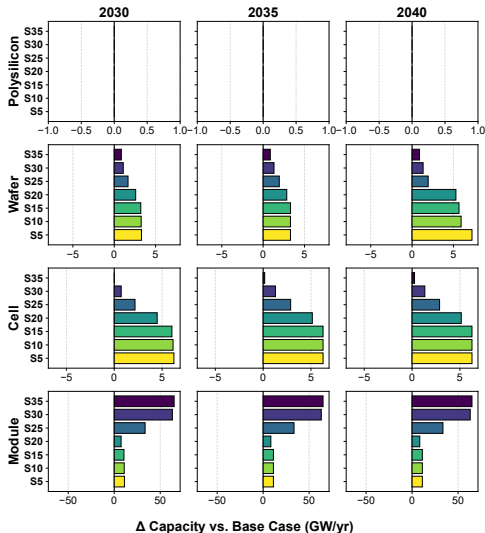
■ Critical Raw Materials Act:

$$\hat{m}_{y,r}^{\text{mine}} + \hat{m}_{y,r}^{\text{rec}} \geq \chi^{\text{crma}} \times D_{y,r}^{\text{mat}} \quad (2)$$

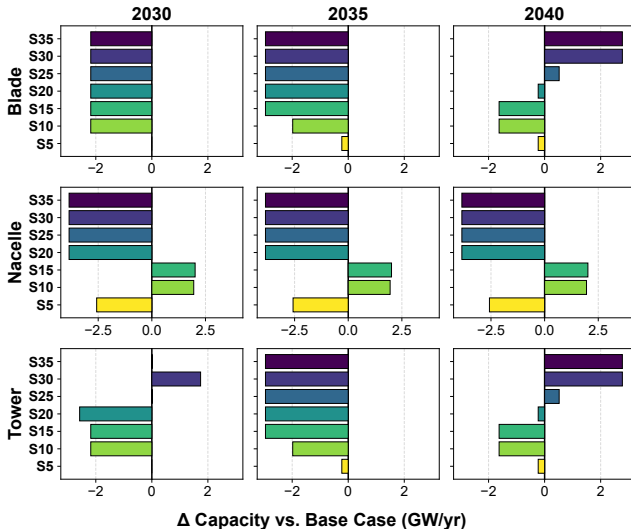
Final Installed Capacities: Combined Relative Change



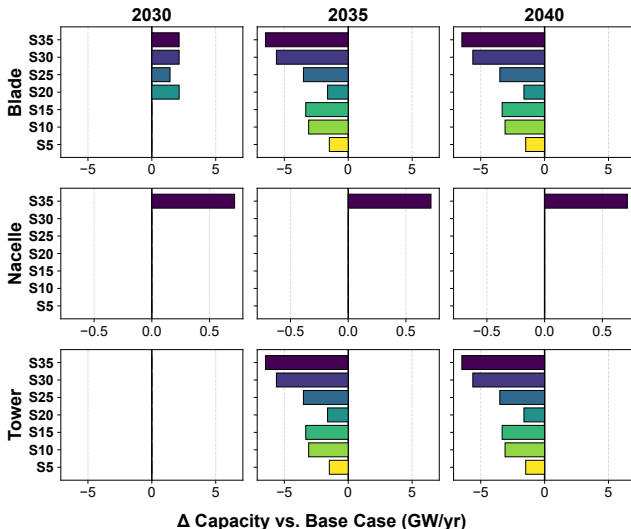
Processing Capacities: Solar PV



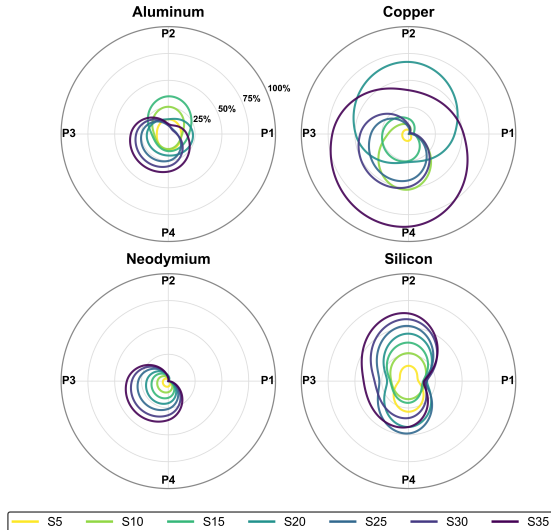
Processing Capacities: Wind Onshore



Processing Capacities: Wind Offshore



CRM Sourcing: Selected Examples



Conclusion

- **Proof of Concept:** The energy-material nexus is successfully endogenized, even under severe data scarcity.
- **Policy Coordination:** Industrial and material targets must be aligned; weak mandates fail to incentivize domestic processing (e.g., Wind).
- **Capacity Patterns:** Cross-technology materials exhibit complex interactions requiring deeper study. Solar PV is the preferred choice in low CRMA scenarios, whereas for wind, the opposite is true.
- **Unexpected Pathways:** The model determines CCUS technology as cost-optimal option for electricity decarbonization.

Future Work & Next Steps

Pathway A: Data Scarcity Resolved

- Obtain exact CRM price trajectories and recycling costs.
- Shift focus to granular geographic modeling (higher solve times).
- Apply methodology to other sectors (e.g., EV batteries).

Pathway B: Data Scarcity Persists

- Use massive scenario runs to map parameter uncertainty.
- Keep model strictly linear to guarantee fast solve times.
- Explore soft-linking to prevent non-linear LP-bottlenecks.

Thank You

for your attention.

Happy to answer your questions!

Find out more about us at: es3m.eu

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